# Digital filters

Practice questions · Day 17 · Mon 9 Nov · pack D10

## 5. Practice

### From your course files

**T8 Q1 (tutorial).** Determine the coefficients of an FIR filter described by the difference equation

$$y(n) = b_0 x(n) + b_1 x(n-1) + b_2 x(n-2)$$

such that

i. It rejects the frequency component $\omega_0 = 2\pi/3$ completely,
ii. Its frequency response is normalised to $H(\omega) = 1$ at $\omega = 0$, and
iii. $b_0 = b_2$ so that it has a linear phase response.

Compute and sketch the magnitude and phase response of the filter to confirm that it satisfies the above requirements.

**T8 Q2 (tutorial).** Compute the magnitude and phase response of the comb filter

$$H(z) = \frac{1}{5} \sum_{k=0}^{4} z^{-k}$$

and determine the frequencies of continuous-time sinusoids that cannot pass through this filter if an ADC with sampling frequency $F_s = 1$ kHz is employed.

**T8 Q3 (tutorial).** A digital notch filter is required to remove an undesirable 50-Hz hum associated with a power supply in an ECG recording application. The sampling frequency used is 500 Hz. Design a second-order FIR notch filter for this purpose with frequency response satisfying $H(\omega) = 1$ at $\omega = \pi$.

**T8 Q4 (tutorial).** Consider the FIR system:

$$y(n) = x(n) - 0.9x(n-4)$$

(a) Sketch its pole-zero plot.
(b) Sketch roughly its magnitude response using the pole-zero plot.
(c) Determine the transfer function of its causal inverse system. Is it stable?
(d) Sketch roughly the magnitude response of the inverse system using the pole-zero plot.

**T8 Q5 (tutorial).** A communication channel is modelled by the transfer function

$$H(z) = \frac{1 + 0.7z^{-1} - 1.2z^{-2}}{1 - 0.8z^{-1} + 0.64z^{-2}}, \qquad \text{ROC: } |z| > 0.8$$

(a) Sketch its pole-zero plot.
(b) Is the channel a minimum-phase, maximum-phase or nonminimum-phase system?
(c) Determine the transfer function and impulse response of its stable inverse. Is it causal, anticausal or noncausal?

## New questions

**N1.** [3 marks] A lowpass FIR filter has $h_{LP}(n) = \{0.25, \underset{\uparrow}{0.5}, 0.25\}$.

a. Find $h_{HP}(n) = (-1)^n h_{LP}(n)$ and $H_{HP}(z)$. [1]
b. Give the zeros of both filters and $|H|$ at $\omega = 0, \pi/2, \pi$ for both. [2]

**N2.** [6 marks] An instrument samples at $F_s = 1$ kHz and picks up 50 Hz mains hum.

a. Design a second-order FIR notch with unity gain at DC. [2]
b. Evaluate its gain at 100 Hz and at 500 Hz. Comment. [1]
c. Add poles at $0.95e^{\pm j\omega_0}$. Find $b_0$ for unity DC gain and write $H(z)$. [2]
d. Evaluate the gain of (c) at 100 Hz and 500 Hz. [1]

**N3.** [4 marks] A moving-average filter $y(n) = \frac{1}{M+1}\sum_{k=0}^{M} x(n-k)$ runs at $F_s = 8$ kHz.

a. Choose $M$ so the filter nulls 1 kHz and all its harmonics up to $F_s/2$. List the null frequencies. [2]
b. Find the gain at 500 Hz and the group delay in ms. [2]

**N4.** [6 marks] A digital resonator at $F_s = 8$ kHz has poles at $0.95e^{\pm j\omega_0}$ with $F_0 = 1$ kHz and zeros at $z = \pm 1$.

a. Write $H(z)$ and the difference equation in terms of $b_0$. [2]
b. Find $b_0$ so that $|H(\omega_0)| = 1$. What are $|H(0)|$ and $|H(\pi)|$? [2]
c. Move the poles onto the unit circle and set $b_0 = \sin\omega_0$. Write the difference equation and the first 4 samples of the impulse response. [2]

**N5.** [7 marks] A channel is $H(z) = \dfrac{1 - 1.5z^{-1} - z^{-2}}{1 - 0.4z^{-1}}$, ROC $|z| > 0.4$.

a. Find the poles and zeros and classify the phase type. [2]
b. Is the causal inverse stable? [1]

c. Find the stable inverse $H_I(z)$, its ROC and $h_I(n)$. Is it causal? [3]
d. Evaluate $h_I(n)$ for $n = -2, \ldots, 2$. [1]

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**N6.** [5 marks] $H(z) = \dfrac{0.5 + z^{-1}}{1 + 0.5z^{-1}}$ (causal).

a. Find the pole and zero. Show $|H(\omega)| = 1$. [2]
b. Is it minimum phase? Does it have a stable causal inverse? [1]
c. Show that the group delay is $\tau(\omega) = \dfrac{1 - a^2}{1 + 2a\cos\omega + a^2}$ with $a = 0.5$, and evaluate it at $\omega = 0, \pi/2, \pi$. [2]

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## 6. Quick recall check

1. An FIR filter has $h(n) = \{2, -1, 3, -1, 2\}$. Is it linear phase, and what is its group delay?
2. A 10-tap moving average runs at $F_s = 2$ kHz. What is its lowest null frequency?
3. Convert $H_{LP}(z) = \dfrac{1 + 2z^{-1} + z^{-2}}{1 - 0.5z^{-1} + 0.2z^{-2}}$ to highpass.
4. A causal stable $H$ has zeros at $0.5$ and $-1.2$. What kind of system is its stable inverse?
5. Where are the poles of a digital oscillator, and why is it not BIBO stable?